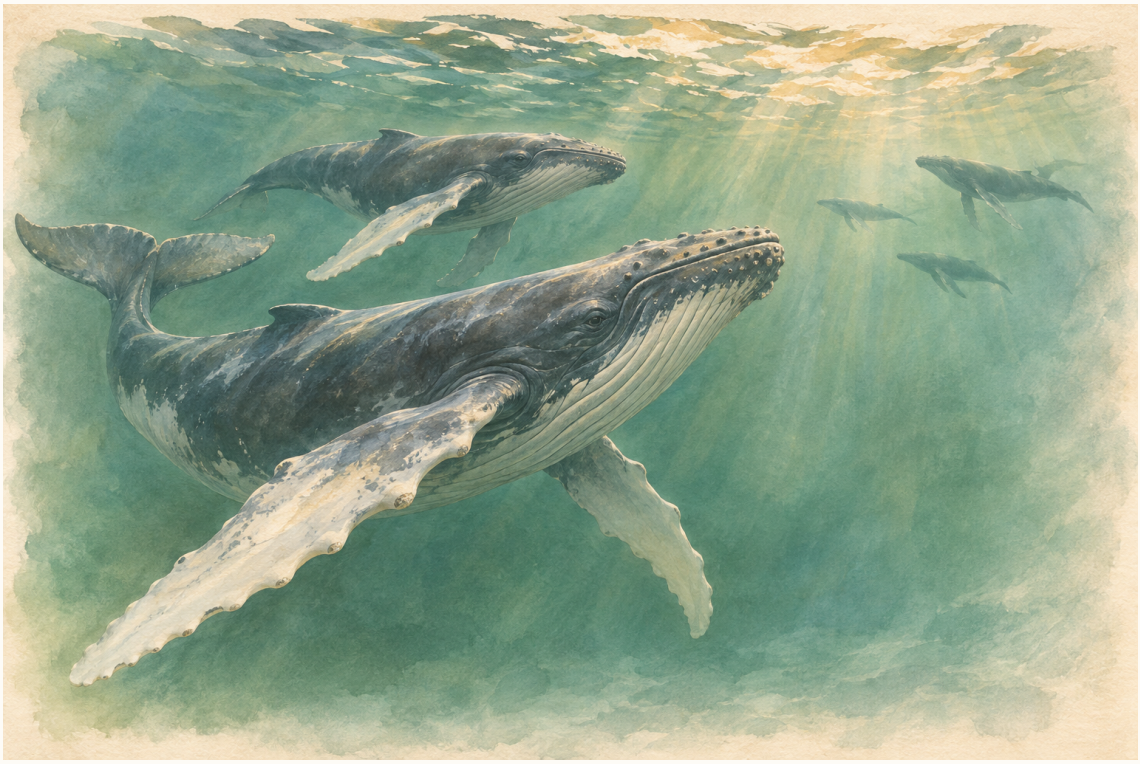
鲸歌易帜

A Change of Anthem in the Deep



新歌如何跨过海洋，传遍鲸群？

英文原文

Each winter, **male humpback whales**, and only males, fill their **breeding grounds** with a **haunting melody**. **Sound units constitute** phrases, and phrases build **themes** in an **elaborate hierarchy**. Each **rendition** lasts minutes, and the song can **recur** for hours, probably for **courtship**. All the males in a **population** sing an **identical** version, which **evolves subtly** each year.

That is the rule. **Off** eastern Australia, it was broken. From 1995 to 1998, **researchers** used hydrophones, or underwater **microphones**, to record humpbacks on their **seasonal migration** along the **coastline**. In the mid-1990s, **merely** two of 82 singers **emitted** a **distinctive** new song, **utterly unlike** the **reigning chorus**.

Within about two years, **virtually** every singer had **converted**, and by 1998 the old song had been **displaced wholesale**. A study published in Nature in November 2000 found that the new song **resembled** that of the Indian Ocean population off western Australia, apparently **ferried** east by **a handful of** visiting **performers**.

Nor was it a **freak**. In April 2011, a study in Current Biology **analysed** over a decade of **recordings** from six South **Pacific** populations. Song types **primarily** arose off eastern Australia, then **rippled eastward** through **successive** populations to French **Polynesia**, **generally** crossing the **region** within about two years.

Such **rapid change**, too fast for genetic **evolution**, points to **imitation** rather than **inheritance**, a **persuasive** case of **cultural transmission**. Why males sing, and why they **overhaul** their songs, remain **controversial**. One idea is that **novelty** itself **spurs** change. Some **scholars** are **reluctant** to **define** this **as** culture in the human sense. Whatever the label, a few **outsiders** off eastern Australia **reshaped** an entire population's music.

中文译文

每年冬天，雄性座头鲸，而且只有雄性，会让它们的繁殖地回荡着一段萦绕于心头的旋律。若干声音单元构成乐句，乐句又组成主题，形成一个复杂精细的层级结构。每一次表演/演奏持续数分钟，而这首歌能反复出现数个小时，很可能是为了求偶。同一种群中的所有雄性都唱着完全相同的版本，而它每年都不易察觉地演变。

这就是规则。在澳大利亚东部海域，这个规则被打破了。1995年至1998年间，研究人员用水听器，即水下麦克风，录下了沿海岸线进行季节性迁徙的座头鲸。20世纪90年代中期，82只歌者中仅有两只发出了一首与众不同的新歌，完全不像占据主导地位的合唱。

大约两年之内，几乎每只歌者都已转变；到1998年，旧歌已被全盘取代。2000年11月发表于《自然》（Nature）的一项研究发现，这首新歌与澳大利亚西部海域印度洋种群的歌相似，似乎是由一小批来访的歌手向东带来的。

这也并非不寻常的事。2011年4月，《当代生物学》（Current Biology）上的一项研究分析了南太平洋六个种群十余年的录音。歌型主要起源于澳大利亚东部海域附近，然后向东如同涟漪般扩散，穿过相继的一个个种群传至法属波利尼西亚，通常在大约两年内横跨该区域。

如此快速的变化，对基因演化来说太快了，指向的是模仿而非遗传，即文化传递的一个有说服力的例证。雄性为何歌唱，以及它们为何要全面改造它们的歌曲，仍然存在争议。一种观点是，新奇本身推动了变化。一些学者不愿把这种现象界定为人类意义上的文化。不管叫什么名号，澳大利亚东部海域的一小批外来者，重塑了整个种群的音乐。

词汇与表达

male adj.
雄性

humpback whales n.
座头鲸

breeding grounds n.
繁殖地

haunting adj.
萦绕于心头的

melody n.
旋律

Sound units n.
声音单元

constitute v.
构成
Sound units constitute phrases,

themes n.
主题

elaborate adj.
复杂精细的
themes in an elaborate hierarchy.

hierarchy n.
层级结构
themes in an elaborate hierarchy.

rendition n.
表演/演奏

recur v.
反复出现
and the song can recur for hours,

courtship n.
求偶
for hours, probably for courtship.

population n.
种群

identical adj.
完全相同的
population sing an identical version,

evolves v.
演变

subtly adv.
不易察觉地

Off prep.
在……海域

researchers n.
研究人员

microphones n.
麦克风

seasonal adj.
季节性

词汇与表达（续）

migration n.
迁徙
on their seasonal migration along the

coastline n.
海岸线

merely adv.
仅

emitted v.
发出了

distinctive adj.
与众不同的
emitted a distinctive new song,

utterly unlike adv.
完全不像

reigning adj.
占据主导地位的

chorus n.
合唱

virtually adv.
几乎
two years, virtually every singer had

converted v.
转变

displaced v.
取代
old song had been displaced wholesale.

wholesale adv.
全盘

a handful of n.
一小批
by a handful of visiting performers.

resembled v.
与……相似
the new song resembled that of the

ferried v.
带来

performers n.
歌手

freak n.
不寻常的事

analysed v.
分析了

recordings n.
录音

Pacific n.
太平洋

primarily adv.
主要
Song types primarily arose off eastern

rippled v.
如同涟漪般扩散

词汇与表达（续）

eastward adv.
向东

successive adj.
相继的
through successive populations to

Polynesia n.
波利尼西亚

generally adv.
通常

region n.
区域

rapid change phr.
快速的变化

evolution n.
演化
too fast for genetic evolution,

imitation n.
模仿
to imitation rather than inheritance,

inheritance n.
遗传
to imitation rather than inheritance,

persuasive adj.
有说服力的

cultural adj.
文化

transmission n.
传递
case of cultural transmission.

overhaul v.
全面改造
why they overhaul their songs, remain

controversial adj.
存在争议
their songs, remain controversial.

novelty n.
新奇
is that novelty itself spurs change.

spurs v.
推动了

scholars n.
学者

reluctant adj.
不愿
Some scholars are reluctant to define

define as v.
界定为

outsiders n.
外来者

reshaped v.
重塑了

文章简述

- 常态：同一群体的雄性座头鲸会唱同一个版本的歌，通常每年只作细微改变，歌唱可能用于求偶。
- 换歌：研究者在澳洲东部录到的82头鲸中，最初仅两头唱全新歌；约两年后，几乎整个鲸群都改唱新歌。
- 来源：新歌与澳洲西部鲸群的歌相似，研究推测是少量来访鲸把它带到了东部。
- 传播：六个南太平洋鲸群的十余年录音显示，新歌多从澳洲东部出现，再逐群向东传播，通常约两年传到法属波利尼西亚。
- 解释：变化快得来不及靠基因一代代改变，支持鲸会相互模仿、学习的解释；为何换歌、是否等同人类文化，仍有争议。